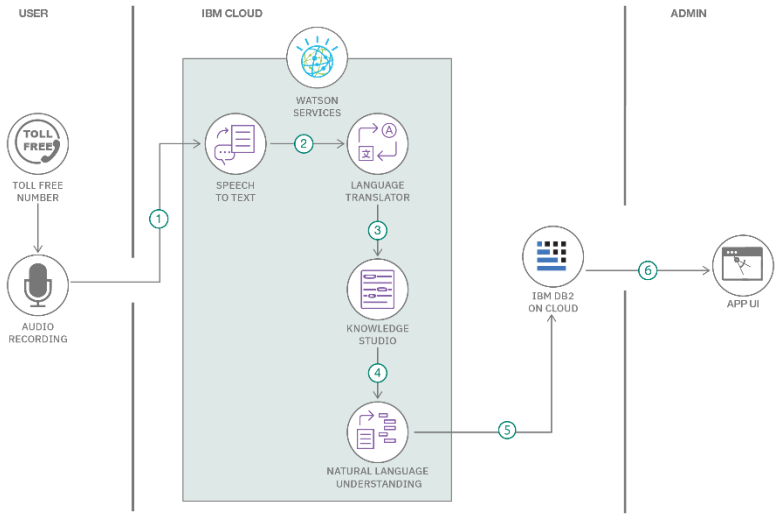
Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	2 March 2026
Team Members Name	Aditi Parashar ,Adarsh Singh ,Aditya
Project Name	**Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau**
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Project: Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau



Guidelines:

Include all the processes (As an application logic / Technology Block)
Provide infrastructural demarcation (Local / Cloud)
Indicate external interfaces (third party API's etc.)
Indicate Data Storage components / services
Indicate interface to machine learning models (if applicable)

S.No	Component	Description	Technology
1.	User Interface	Tableau Desktop and Tableau Public dashboards serve as the user interface. Users interact through interactive visual dashboards, story slides, filters, and tooltips. Tableau Desktop, Tableau Public (Web UI)	Tableau Desktop 2022, Tableau Public
2.	Data Cleaning & Preparation	Handles missing values, standardizes column formats, creates calculated fields (Sale_Price bin, Age Since Last Renovation, Ever Renovated flag) to prepare the dataset for Tableau analysis.	Java / Python
3.	Tableau Data Connection	Connects the cleaned CSV dataset to Tableau, configures data types, sets up relationships, and enables dynamic filtering using dimensions and measures across all worksheets.	Tableau Data Source Connection (CSV)
4.	Calculated Fields & Binning	Builds calculated fields and bins in Tableau such as Sale_Price (bin), AVG(Sale_Price), CNT(Sale_Price), and Age of House groups to power all dashboard visualizations.	Tableau Calculated Fields & Bins
5.	Database	Structured flat-file dataset (CSV) containing 21,609 housing records with columns for sale price, bedrooms, bathrooms, floors, basement area, house age, and renovation status.	CSV Flat File (Transformed_Housing_Data2.csv)
6.	Cloud Database	Tableau Public cloud hosting allows the published dashboard to be accessed online by real estate agents, investors, and buyers without requiring local Tableau installation.	Tableau Public (Cloud Hosting)
7.	File Storage	Local file system storage for the raw and transformed CSV dataset and Tableau workbook (.twbx) file containing all dashboards, worksheets, and story slides.	Local Filesystem (.csv, .twbx files)
8.	Tableau Data Extract API	Used to extract, refresh and manage data connections between the CSV source and Tableau worksheets for dashboard updates.	Tableau Data Extract API (TDE)
9.	External API (Not Used)	Not applicable for this project. No secondary external API is used as the solution relies entirely on local CSV data processed within Tableau.	N/A

10.	Statistical Analysis Model	Tableau built-in aggregation functions (AVG, COUNT, SUM) are used to compute key statistics such as average sale price, property count, and price distribution across all price bins and house age groups.	Tableau Aggregation Functions (AVG, COUNT, SUM, BIN)
11.	Infrastructure (Server / Cloud)	The solution is developed and run locally on a Windows PC using Tableau Desktop. Dashboards are published to Tableau Public for online access by stakeholders. Local: Windows PC with Tableau Desktop installed Cloud: Tableau Public (free cloud hosting for published dashboards)	Local (Tableau Desktop) + Tableau Public (Cloud)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Tableau Public is a free, open-access platform used to publish and share interactive dashboards. The CSV dataset is processed using open data formats compatible with all major analytics tools.	Tableau Public, CSV (Open Data Format)
2.	Security Implementations	The dataset contains no personally identifiable information (PII). Tableau Public dashboards are published with view-only access. Data integrity is maintained through transformation controls during the cleaning phase.	View-only Access Control (Tableau Public), No PII Data
3.	Scalable Architecture	The solution follows a 3-layer architecture: Data Layer (CSV), Processing Layer (Tableau Desktop), and Presentation Layer (Dashboards/Story). New datasets from additional cities or years can be integrated without redesigning the dashboard structure.	3-Layer Architecture: Data (CSV) → Processing (Tableau Desktop) → Presentation (Dashboard)
4.	Availability	Dashboards published on Tableau Public are available 24/7 via any web browser without requiring Tableau Desktop installation. Users can access and interact with all visualizations online at any time.	Tableau Public (Always-on Cloud Hosting)
5.	Performance	Tableau's in-memory data engine processes 21,609 records efficiently, rendering all dashboards and story	Tableau Data Engine (In-Memory Processing), Tableau Extracts (.hyper files)

S.No	Characteristics	Description	Technology
		slides with fast response times. Data extracts (.hyper files) are used to optimize query performance and reduce load times.	